

BUILT FOR THE NEXT GENERATION OF POWER.

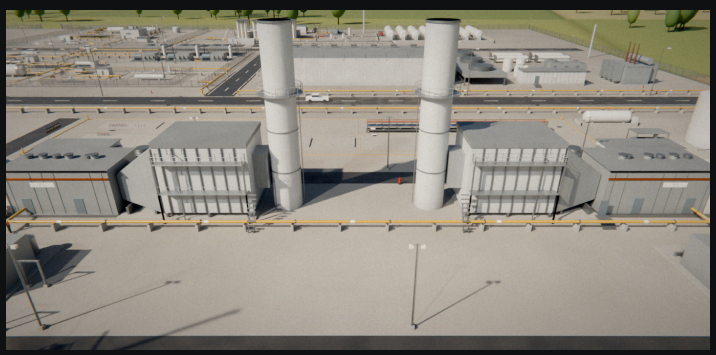
The power plant has changed. One gas plant selling megawatt-hours has become a campus: efficient combined cycle, fast modular power, battery storage, carbon capture, green hydrogen and the data centres that now drive demand. Here is what changed, and the cable that makes it work.

01 — WHAT CHANGED

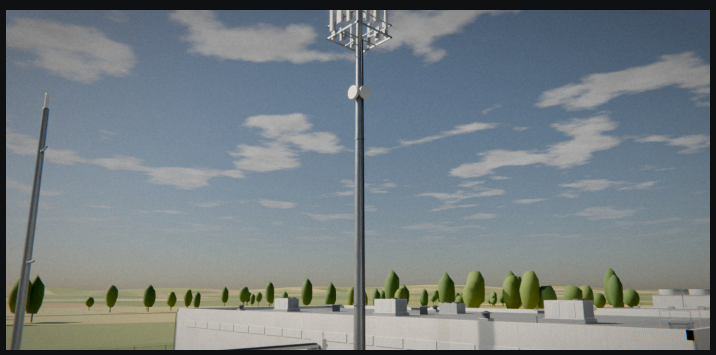
THEN AND NOW.

Plants built in the last decades were designed to run steady and sell energy. The plant of the next generation has to start fast, follow renewables, cut its emissions and power new loads, and it has to be built faster. Every one of those changes runs on cable.

	THEN	NOW: SW-3X1
THE PLANT	One gas plant, one product: MWh	A campus: ~1,350 MW combined cycle, ~339 MW modular power, battery storage, a data centre
FLEXIBILITY	Baseload, slow starts	RICE engines and aeroderivatives on in minutes; batteries absorb the swings
EMISSIONS	Treated as a cost of running	Carbon capture on the HRSG exhaust; hydrogen blended into the gas turbines
FUEL	Natural gas only	Gas today, green hydrogen from a 25 MW electrolyser on site, hydrogen-ready turbines
CABLE INSTALL	Pulled piece by piece, on site	Prefab cable modules, pre-terminated MV, pads pre-wired before the equipment arrives
SET-UP	Reel jacks, axles and a crew	SIMpull Reels, BARRELs and CoilPAKs one person moves; Maxis pullers and feeders
CONNECTIVITY	Copper and paper, manual rounds	Fibre backbone, private 5G / LTE, licence-plate cameras, access control
DESIGN	2D drawings, field-routed cable	One 3D model: every circuit routed, sized and scheduled before the first pull



Simple-cycle pair, stacks together and inlets outboard: fast-start power beside the combined cycle.



Private 5G / LTE monopole: the plant network for cameras, sensors and crews.

stephan.hardt@southwire.com

The power block with its cable runs highlighted in copper.



Fast-start simple-cycle pair in the modular yard.

ONE CAMPUS. MANY TECHNOLOGIES.

SW-3X1 pairs an efficient 3x1 H-class combined cycle with modular power that starts in minutes, battery storage that absorbs the swings, and the behind-the-meter data centre that is today's fastest-growing load. Each one is a different cable job, from 230 kV to Cat 6A.

- **COMBINED CYCLE** 3 H-class gas turbines, 3 HRSGs, 1 steam turbine: ~1,350 MW of efficient baseload
- **MODULAR POWER** RICE engines 144 MW, simple cycle 100 MW, mobile turbines 70 MW, gensets, fuel cells
- **BATTERY STORAGE** ~60 MW / 120 MWh to smooth renewables and bridge starts
- **DATA CENTRE** ~150 MW of IT load fed behind the meter from the modular yard
- **RESERVED FOR GROWTH** SC-3 / SC-4 pad poured and pre-wired: phase two plugs in
- **THE CABLE** HV, MV-105, TC-ER, ARMOR-X MC-HL, DC, instrumentation, fibre: one schedule, one supplier

stephan.hardt@southwire.com

Carbon capture: absorbers A-C, strippers and compression beyond.



Electrolyser hall: PEM skids, rectifiers and the DC cable trays.

LOWER CARBON, BUILT IN.

Carbon capture takes CO2 out of the exhaust before it reaches the stack. A 25 MW PEM electrolyser makes green hydrogen on site from renewable or surplus power, and the gas turbines are ready to burn it blended with natural gas. Each system brings its own cable: hazardous-area, high-current DC and dense instrumentation.

- **CARBON CAPTURE** amine absorbers on the HRSG exhaust, strippers, CO2 compression for transport
- **GREEN HYDROGEN** 25 MW PEM electrolysis, ~10.8 t / day, rectifiers and DC buses
- **HYDROGEN-READY** receiving station, blending skid and hydrogen-capable turbines
- **HAZARDOUS AREAS** ARMOR-X MC-HL power and control where gas and hydrogen are handled
- **HIGH-CURRENT DC** 2 kV DC conductors from rectifiers to stacks on dedicated tray
- **INSTRUMENTATION** shielded PLTC / ITC for the analysers, ESD and safety systems

stephan.hardt@southwire.com

SC-4 reserved pad: prefab cable modules installed, pigtails waiting at each landing point.



Office fit-out: MC, Cat 6A on J-hooks, SIMpull and Maxis on the floor.

WIRED BEFORE IT ARRIVES.

The modern plant is built in parallel. Cable systems are made in the factory, tested, and set on pads that are pre-wired before the equipment lands; inside, the crews pull with tools that take the jacks, lube and guesswork out of the job. On SW-3X1 the SC-4 pad shows it: everything but the turbine is in.

- **PREFAB CABLE MODULES** 40 ft factory modules: power, control and fibre tiers laid, tied and tested
- **PLUG-AND-PLAY PADS** pigtails coiled and tagged at every landing: set, connect, energise
- **SIMPULL** cut-to-length, paralleled reels and BARRELs, no lube, factory pulling heads
- **MAXIS** XD10 pullers and feeders with tension monitoring and Triggers safety switch
- **DIGITAL MODEL** every circuit routed and sized in one model; the schedule and BOM come from it
- **FIT-OUT** MC branch circuits, Cat 6A, FPLR fire alarm and fibre, from the same supplier

stephan.hardt@southwire.com

05 — SUSTAINABILITY

FROM THE TRACK TO THE TURBINE.

Sustainability at Southwire is a commitment, not a campaign: making cable responsibly, helping customers build with less waste, and putting electrification to work wherever it is needed, from the grid to the race track.

FORMULA E PHOTO + LOGO
INSERT AFTER BRAND APPROVAL

- **FORMULA E** Southwire is Formula E's Official Wire and Cable Provider, announced in March 2023 for the 2022/23 season.
- **NET-ZERO RACING** Formula E describes itself as carbon neutral since its first season in 2014, and as the most sustainable sport (Global Sustainability Benchmark in Sports, 2023).
- **SAME KNOW-HOW** The electrification that powers an all-electric race series is the same that powers batteries, electrolysers and the cable runs of a modern plant.
- **LESS WASTE ON SITE** Cut-to-length reels, returnable reels and payoffs, no pulling lube, fewer reels to handle and dispose of.
- **BUILT IN THE PLANT** Prefab and pre-wired pads move work into the factory: fewer hours on site, safer crews, less rework.
- **SOUTHWIRE SUSTAINABILITY** The company's sustainability commitments: see southwire.com for the current programme.

SOUTHWIRE: WIRING THE ENERGY TRANSITION.

Plant images and figures from the SW-3X1 conceptual 3D model: illustration, not engineered, ratings approximate. Formula E facts as published by Southwire and Formula E; sustainability claims are the series' own.

stephan.hardt@southwire.com